The Verification Tax: Fundamental Limits of AI Auditing in the Rare-Error Regime

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Abstract

As AI models become more capable, their error rates decrease — yet we show that verifying their reliability becomes fundamentally *harder*, not easier. We establish minimax-optimal rates for estimating the Expected Calibration Error (ECE) of classifiers with error rate ε, revealing a "verification tax" that grows as errors become rare. Our Le Cam lower bound shows that any ECE estimator incurs error $\Omega(\sqrt{\varepsilon/m})$, and a reduction to L_1 functional estimation yields the nonparametric minimax rate $\Theta((L\varepsilon/m)^{1/3})$ up to logarithmic factors, which histogram binning with an ε-adaptive bin count achieves. A phase transition emerges at $m \sim 1/\varepsilon$: below this threshold, even detecting miscalibration is impossible. We prove that white-box model access (inspecting weights, architecture, training data) provides no fundamental advantage unless it identifies a low-dimensional parametric family for the calibration function — in which case the rate improves to the parametric $O(1/\sqrt{m})$. We validate these bounds empirically via synthetic experiments with known ground truth, and illustrate the practical implications on representative model configurations. Our results formalize the tension between AI capability and auditability, with direct implications for regulatory frameworks that mandate calibration verification.

1 Introduction

The base rate problem — that rare conditions are hard to detect — is among the oldest results in statistics. A screening test for a disease with 1% prevalence requires thousands of patients to achieve reliable sensitivity, no matter how accurate the test. As prevalence falls, the required sample size grows without bound. This is not a limitation of screening protocols. It is a consequence of Bayes' theorem.

We show that AI auditing faces the same fundamental barrier. The "disease" is miscalibration — the gap between a model's stated confidence and its actual accuracy. As models improve and errors become rare, miscalibration becomes a rare condition, and detecting it requires an ever-growing quantity of labeled evaluation data. We call this the *verification tax*: the statistical cost that must be paid to confirm that an AI system is as reliable as it claims to be.

The verification tax is not a limitation of current evaluation methods. It is a law. No algorithm, no amount of compute, no access to model internals can circumvent it. The tax applies universally across calibration, fairness, and robustness verification. It compounds exponentially through multi-component pipelines. And it creates a *verification horizon* — a critical model capability beyond which the world's supply of labeled data is insufficient to verify performance.

But the tax is not without relief. Adaptive evaluation protocols — where the auditor chooses which inputs to test — achieve a strictly faster verification rate, eliminating the dependence on calibration smoothness entirely. Parametric assumptions (e.g., temperature

scaling) reduce the tax to the standard parametric rate. And prior verification transfers across model updates proportionally to the magnitude of the update.

Our contributions are as follows:

1. **The verification tax.** We prove that estimating the expected calibration error (ECE) for a model with error rate ε requires $m = \Omega((L\varepsilon/\delta^3))$ labeled samples to achieve accuracy δ, where L is the Lipschitz constant of the calibration function (Theorems 7–9). The lower and upper bounds match up to logarithmic factors. This rate reveals a phase transition at $m \sim 1/\varepsilon$: below this threshold, even detecting miscalibration is impossible.
2. **Universality.** The verification tax applies identically to group fairness verification (with effective sample size scaled by group proportion) and robustness verification (with error rate under perturbation), unifying three distinct evaluation problems under a single framework (Theorem 13).
3. **Impossibility results.** White-box access to model weights, architecture, and training data does not reduce the verification tax unless it identifies a parametric calibration family (Theorem 16). Self-verification without external labels is impossible regardless of model access or sample size (Theorem 25). Strategic miscalibration cannot increase the tax beyond the standard rate (Proposition 18).
4. **Adaptive evaluation.** When the auditor can choose which confidence levels to query, the minimax verification rate improves to $\Theta(\sqrt{\varepsilon/m})$, eliminating the Lipschitz constant entirely (Theorem 20). This is tight: the Le Cam lower bound matches the two-phase upper bound with no gap.
5. **Scaling consequences.** Under standard scaling laws, there exists a verification horizon — a critical model size beyond which verification is infeasible given finite labeled data (Theorem 33). Iterative recalibration hits a ceiling of $K^* = O(\log M_{\text{total}})$ verifiable rounds (Proposition 39). Compositional AI systems face verification costs that grow exponentially with pipeline depth (Theorem 47).
6. **Temporal decay.** Every verification has a half-life determined by the deployment distribution's drift rate (Theorem 43), ranging from hours in financial applications to months in healthcare.
7. **Empirical validation.** We validate the theoretical predictions via synthetic experiments with known ground truth, confirming the predicted scaling exponents, phase transition location, and active-passive gap (Section 4).

The remainder of this paper is organized as follows. Section 2 introduces the formal framework and establishes the core verification tax (Theorems 7–9) along with its generalization to fairness and robustness verification. Section 3 presents the impossibility results, adaptive and sequential verification strategies, and parametric and transfer reductions. Section 5 translates the theory into practical formulas for holdout sizing, stratified evaluation, and verification budgeting. Section 6 derives the verification horizon, dynamics, recalibration trap, and half-life. Section 7 establishes the compositional verification tax for multi-component AI systems. Section 8 discusses related work, and Section 9 concludes with limitations and directions for future work.

2 Setup and Core Theory

2.1 Formal Setup

Definition 1 (Binned Expected Calibration Error)**.** Let $f : \mathcal{X} \to \mathcal{Y} \times [0, 1]$ be a classifier outputting (prediction, confidence). Let $p(x) = f(x)_{\text{conf}}$ denote the confidence score. Partition $[0, 1]$ into B equal-width bins $I_1, \ldots, I_B$ where $I_b = ((b-1)/B,\, b/B]$. The binned ECE is:

$$\text{ECE}_B = \sum_{b=1}^{B} \frac{n_b}{n} \left|\text{acc}(I_b) - \text{conf}(I_b)\right| \tag{1}$$

where $n_b = |\{i : p(x_i) \in I_b\}|$, $\mathrm{acc}(I_b) = \frac{1}{n_b}\sum_{i:p(x_i)\in I_b} \mathbb{1}[y_i = \hat{y}_i]$, and $\mathrm{conf}(I_b) = \frac{1}{n_b}\sum_{i:p(x_i)\in I_b} p(x_i)$.

Primary metric throughout this paper: Top-label binned ECE with uniform-width binning.

Definition 2 (Calibration Gap Function)**.**

$$\Delta(p) = \eta(p) - p, \quad \text{where } \eta(p) = \mathbb{E}[Y = \hat{y} \mid p(X) = p] \tag{2}$$

So $\mathrm{ECE}_B = \sum_b \frac{n_b}{n}|\bar{\Delta}_b|$ where $\bar{\Delta}_b$ is the average calibration gap in bin b.

Definition 3 (Error Rate)**.**

$$\varepsilon = \Pr[Y \neq \hat{y}] = \mathbb{E}[1 - \eta(p(X))] \tag{3}$$

For a well-calibrated model, $\varepsilon = \mathbb{E}[1 - p(X)]$. In the rare-error regime, $\varepsilon \to 0$, and label variance near high-confidence predictions satisfies $\mathrm{Var}(Y \mid p(X) \approx 1 - \varepsilon) = \varepsilon(1 - \varepsilon) \approx \varepsilon$.

Definition 4 (Lipschitz Calibration Function Class)**.**

$$\mathcal{F}(L) = \{\Delta : [0, 1] \to [-1, 1] \mid |\Delta(p_1) - \Delta(p_2)| \leq L|p_1 - p_2| \; \forall\, p_1, p_2\} \tag{4}$$

Definition 5 (Minimax ECE Estimation Risk)**.**

$$R^*(m, \varepsilon, L) = \inf_{\widehat{\mathrm{ECE}}} \sup_{\Delta\in\mathcal{F}(L)} \mathbb{E}\left[\left|\widehat{\mathrm{ECE}} - \mathrm{ECE}(\Delta)\right|\right] \tag{5}$$

where the infimum is over all estimators based on m i.i.d. labeled samples from the classifier.

Lemma 6 (Bridge Lemma: Binned ECE $\leftrightarrow$ Smooth CE)**.** *Let* $\mathrm{CE}_{\mathrm{smooth}} = \int |\Delta(p)|\, d\mu(p)$. *Then for B equal-width bins:*

$$\mathrm{ECE}_B \leq \mathrm{CE}_{\mathrm{smooth}} \leq \mathrm{ECE}_B + \frac{L}{B} \tag{6}$$

Proof. **Lower bound:** By Jensen's inequality within each bin, $|\mathbb{E}[\Delta(p) \mid p \in I_b]| \leq \mathbb{E}[|\Delta(p)| \mid p \in I_b]$, so $\mathrm{ECE}_B \leq \mathrm{CE}_{\mathrm{smooth}}$.

Upper bound: Within bin I_b of width $1/B$, Δ varies by at most L/B (Lipschitz). So $|\Delta(p)| \leq |\bar{\Delta}_b| + L/B$ for all $p \in I_b$. Integrating: $\mathrm{CE}_{\mathrm{smooth}} \leq \mathrm{ECE}_B + L/B$. □

2.2 Le Cam Lower Bound

Theorem 7 (Le Cam Lower Bound)**.** *For any $L > 0$ and $\varepsilon \in (0, 1/2)$,*

$$R^*(m, \varepsilon, L) \geq c_1\sqrt{\frac{\varepsilon}{m}} \tag{7}$$

where $c_1 > 0$ is a universal constant. In particular, $c_1 \geq 1/(4\sqrt{2}) \approx 0.177$.

Proof. Construct two hypotheses. Fix a classifier whose score distribution places all mass at $p_0 = 1 - \varepsilon$.

- P_0: $Y \mid p(X) = p_0 \sim \mathrm{Bern}(1 - \varepsilon)$. Calibrated: $\eta(p_0) = p_0$, so $\mathrm{ECE} = 0$.
- P_1: $Y \mid p(X) = p_0 \sim \mathrm{Bern}(1 - \varepsilon - \delta)$. Miscalibrated: $\mathrm{ECE} = \delta$.

Both are consistent with $\Delta \in \mathcal{F}(L)$ for any $\delta \leq L$ (the calibration gap is constant, trivially Lipschitz).

The KL divergence of m i.i.d. samples is:

$$\mathrm{KL}(P_0^m \| P_1^m) = m \cdot \mathrm{KL}(\mathrm{Bern}(1 - \varepsilon) \| \mathrm{Bern}(1 - \varepsilon - \delta)) = \frac{m\delta^2}{\varepsilon(1 - \varepsilon)} \leq \frac{m\delta^2}{\varepsilon} \tag{8}$$

where the last inequality uses $1 - \varepsilon \leq 1$ and the KL approximation $\mathrm{KL}(\mathrm{Bern}(p) \| \mathrm{Bern}(p + \delta)) \approx \delta^2/(p(1-p))$ for small δ.

By Le Cam's lemma (Tsybakov, 2009):

$$R^* \geq \frac{\delta}{2}\left(1 - \mathrm{TV}(P_0^m, P_1^m)\right) \geq \frac{\delta}{2} e^{-\mathrm{KL}(P_0^m \| P_1^m)} \tag{9}$$

where the second inequality uses Bretagnolle–Huber. Set KL = 1, i.e., $\delta = \sqrt{\varepsilon/m}$:

$$R^* \geq \frac{1}{2e}\sqrt{\frac{\varepsilon}{m}} \approx 0.184\sqrt{\frac{\varepsilon}{m}} \tag{10}$$

□

2.3 Minimax Lower Bound via Reduction to L_1 Functional Estimation

Theorem 8 (Minimax Lower Bound)**.** *For the class of L-Lipschitz calibration functions with error rate ε, the minimax ECE estimation error satisfies:*

$$R^*(m, \varepsilon, L) \geq c_2 \left(\frac{L\varepsilon}{m}\right)^{1/3} \cdot (\log m)^{-c_3} \tag{11}$$

where $c_2, c_3 > 0$ are universal constants.

Proof sketch (full proof in Appendix A). **Step 1: Reduction to nonparametric regression.** Within a window of width w around $p = 1 - \varepsilon$, where the score distribution concentrates, the calibration verification problem reduces to estimating the L_1 norm of a regression function. For $p \in [1 - \varepsilon - w/2, 1 - \varepsilon + w/2]$, the labels Y_i given $p(X_i) = p$ are $\mathrm{Bern}(\eta(p)) = \mathrm{Bern}(p + \Delta(p))$, with variance $\eta(p)(1 - \eta(p)) \approx \varepsilon$.

Step 2: Identification with L_1 functional estimation in white noise. By the asymptotic equivalence of nonparametric regression and Gaussian white noise (Brown & Low, 1996; Nussbaum, 1996), estimating $\int |\Delta(p)| dp$ from m observations with per-observation variance ε is asymptotically equivalent to estimating $\int |f(t)| dt$ in the white noise model with effective sample size $n_{\mathrm{eff}} = m/\varepsilon$.

Regularity conditions for asymptotic equivalence. The Brown–Low equivalence (Brown & Low, 1996) requires: (i) the score distribution μ has a density bounded away from zero on the window $[1 - \varepsilon - w, 1 - \varepsilon + w]$, i.e., $\inf_{p \in \mathrm{window}} d\mu/dp \geq c > 0$; (ii) the number of samples m is sufficiently large relative to the smoothness class ($m \geq C(L, w)$ for a constant depending on the Lipschitz constant and window width); and (iii) the Bernoulli noise variance $\varepsilon(1 - \varepsilon)$ is bounded away from zero, which holds whenever $\varepsilon \in (\eta_0, 1 - \eta_0)$ for any $\eta_0 > 0$. Condition (i) is ensured by the score-mass concentration assumption (stated below Theorem 8); conditions (ii) and (iii) are mild for the rare-error regime with $\varepsilon \in (0, 1/2)$ and $m \to \infty$.

Step 3: Application of Lepski, Nemirovski & Spokoiny (1999). By Theorem 2.2 of Lepski et al. (1999), the minimax rate for estimating the L_1 norm of a β-Hölder function in white noise with sample size n is $R^* \geq c \cdot n^{-\beta/(2\beta+1)}/(\log n)^{c'}$. For $\beta = 1$ (Lipschitz) and $n = m/\varepsilon$, incorporating the Lipschitz constant L:

$$R^*(m, \varepsilon, L) \geq c_2 \left(\frac{L\varepsilon}{m}\right)^{1/3} / (\log m)^{c_3} \tag{12}$$

□

2.4 Matched Upper Bound

Theorem 9 (Matched Upper Bound)**.** *Histogram binning with $B^* = \lfloor (L^2 m/\varepsilon)^{1/3} \rfloor$ bins achieves:*

$$\mathbb{E}\left[\left|\widehat{\mathrm{ECE}}_{B^*} - \mathrm{ECE}\right|\right] \leq C \left(\frac{L\varepsilon}{m}\right)^{1/3} \tag{13}$$

where $C > 0$ depends only on the score distribution μ.

Proof. **Bias-variance decomposition.** With B equal-width bins and m total samples:

1. Bias (approximation error). Within each bin of width $1/B$, the calibration gap Δ varies by at most L/B (Lipschitz). By the Bridge Lemma (6):

$$|\mathrm{ECE}_B - \mathrm{CE}_{\mathrm{smooth}}| \leq \frac{L}{B} \tag{14}$$

2. Variance (estimation error). Each bin b has $n_b \approx m/B$ samples (assuming roughly uniform score distribution). The empirical calibration gap $\hat{\Delta}_b$ has:

$$\mathrm{Var}(\hat{\Delta}_b) = \frac{\eta_b(1-\eta_b)}{n_b} \approx \frac{\varepsilon}{m/B} = \frac{\varepsilon B}{m} \tag{15}$$

The estimation error of the full ECE is:

$$\mathbb{E}\left[|\widehat{\mathrm{ECE}}_B - \mathrm{ECE}_B|\right] \leq \sqrt{\frac{\varepsilon B}{m}} \tag{16}$$

using Jensen's inequality and the independence across bins for the dominant term.

3. Total error:

$$\mathbb{E}\left[|\widehat{\mathrm{ECE}}_B - \mathrm{CE}_{\mathrm{smooth}}|\right] \leq \frac{L}{B} + \sqrt{\frac{\varepsilon B}{m}} \tag{17}$$

4. Optimization over B. Set $L/B = \sqrt{\varepsilon B/m}$, solve:

$$\frac{L^2}{B^2} = \frac{\varepsilon B}{m} \implies B^3 = \frac{L^2 m}{\varepsilon} \implies B^* = \left(\frac{L^2 m}{\varepsilon}\right)^{1/3} \tag{18}$$

5. Substituting back:

$$\mathrm{Error} = \frac{L}{B^*} = L \cdot \left(\frac{\varepsilon}{L^2 m}\right)^{1/3} = \left(\frac{L\varepsilon}{m}\right)^{1/3} \tag{19}$$

□

The lower and upper bounds match in the polynomial rate $(L\varepsilon/m)^{1/3}$; the logarithmic gap is inherited from the open problem of L_1 functional estimation (Lepski et al., 1999).

Corollary 10 (Phase Transition)**.** *Holdout-based calibration assessment (multi-bin) beats the trivial calibrator $\hat{c}(p) = 1 - \hat{\varepsilon}$ (single-bin) if and only if $m \gtrsim 1/\varepsilon$. When $\sqrt{\varepsilon/m} > \varepsilon$, i.e., $m < 1/\varepsilon$, even detecting miscalibration is impossible.*

2.5 Generalization Beyond Calibration

The preceding results concern calibration (ECE), but the verification tax phenomenon is not specific to calibration. Any model property that takes the form of an integrated absolute deviation — including group fairness and distributional robustness — faces the same fundamental sample complexity barrier. We unify these under a single framework.

Definition 11 (Verification Functional)**.** A *verification functional* is any quantity of the form:

$$V(\Delta, \mu_G) = \int |\Delta(p)|\, d\mu_G(p) \tag{20}$$

where $\Delta : [0,1] \to \mathbb{R}$ is an L-Lipschitz deviation function and μ_G is a (possibly reweighted) score distribution over a subpopulation or domain G.

Remark 12 (Instantiations)**.** Three important special cases: (i) *Calibration*: G is the full population, $\Delta(p) = \eta(p) - p$, and $V = \mathrm{ECE}$. (ii) *Group fairness*: for demographic group A with proportion π_A, μ_A is the conditional score distribution given group A, and the group-specific ECE is $V(\Delta, \mu_A)$ with effective sample size $m_{\mathrm{eff}} = m\pi_A$. (iii) *Robustness*: under input perturbation $x \to x + \nu$, the model produces shifted scores $p'(x)$ with shifted calibration gap Δ', and the robustness verification functional is $V(\Delta', \mu')$.

Theorem 13 (General Verification Tax)**.** *For any verification functional V with L-Lipschitz integrand, effective noise variance σ^2 (the label variance in the relevant subpopulation), and effective sample size m_{eff}:*

(i) Le Cam bound:

$$R^*(m_{\text{eff}}, \sigma^2, L) \geq c\sqrt{\frac{\sigma^2}{m_{\text{eff}}}} \tag{21}$$

(ii) Minimax rate:

$$R^*(m_{\text{eff}}, \sigma^2, L) = \Theta\left(\left(\frac{L\sigma^2}{m_{\text{eff}}}\right)^{1/3}\right) \quad \textit{up to logarithmic factors} \tag{22}$$

(iii) Active rate:

$$R^*_{\text{active}}(m_{\text{eff}}, \sigma^2, L) = \Theta\left(\sqrt{\frac{\sigma^2}{m_{\text{eff}}}}\right) \tag{23}$$

Proof. The proofs of Theorems 7–9 and 20 use only three properties of the estimation problem: (a) the integrand is L-Lipschitz, (b) the observations are Bernoulli with variance σ^2, and (c) the functional is an L_1 norm. All three verification functionals share this structure with ε replaced by σ^2 and m replaced by m_{eff}. □

Corollary 14 (Fairness Verification Tax)**.** *Verifying equalized calibration (equal ECE across groups) for a demographic group with proportion π requires:*

$$m \geq \frac{1}{\pi} \cdot \frac{L\varepsilon}{\delta^3} \tag{24}$$

For a 5% minority group, this is $20\times$ the cost of verifying the full population. The fairness tax and the verification tax multiply.

Corollary 15 (Robustness Verification Tax)**.** *Verifying calibration under input perturbation requires:*

$$m \geq \frac{L\varepsilon_{\text{pert}}}{\delta^3} \tag{25}$$

where $\varepsilon_{\text{pert}}$ is the error rate under the perturbed distribution. Since perturbation typically increases error rate ($\varepsilon_{\text{pert}} > \varepsilon$), robustness verification is strictly more expensive than standard calibration verification.

3 Extensions

3.1 White-Box Irrelevance

Theorem 16 (White-Box Irrelevance)**.** *Let $\mathcal{S} = \sigma(\theta, A, D_{\text{train}})$ be the sigma-algebra generated by model weights θ, architecture A, and training data D_{train}. For any estimator that is $\mathcal{S}$-measurable and uses m fresh labeled samples, the minimax ECE over $\mathcal{F}(L)$ satisfies:*

$$R^*_{\mathcal{S}}(m, \varepsilon, L) \geq c\sqrt{\frac{\varepsilon}{m}} \tag{26}$$

That is, white-box access does not improve the minimax rate beyond the black-box Le Cam bound (Theorem 7).

Proof. The proof constructs two worlds with the *identical model* but different label distributions.

Fix a model — same weights θ, same architecture A, same training data D_{train}. This model assigns confidence scores $p(x)$ to inputs. The sigma-algebra $\mathcal{S} = \sigma(\theta, A, D_{\text{train}})$ is identical

under both hypotheses. Any $\mathcal{S}$-measurable estimator sees exactly the same model internals under both.

Define two worlds:

- P_0: $Y \mid p(X) = p \sim \mathrm{Bern}(p)$. The model is perfectly calibrated. $\mathrm{ECE} = 0$.
- P_1: $Y \mid p(X) = p \sim \mathrm{Bern}(p - \delta)$ for p near $1 - \varepsilon$. The model is miscalibrated by δ. $\mathrm{ECE} = \delta$.

The model is identical in both worlds — same weights, same predictions, same confidence scores. The only difference is whether the labels agree with the confidences at rate p (world P_0) or at rate $p - \delta$ (world P_1).

An $\mathcal{S}$-measurable estimator receives no additional information for distinguishing P_0 from P_1, because $\mathcal{S}$ is identical under both. The only distinguishing information comes from the m fresh labeled samples.

Formally: conditioned on $\mathcal{S}$ and on the test inputs $x_1, \ldots, x_m$ (which determine scores $p_1, \ldots, p_m$), the labels $y_1, \ldots, y_m$ are independent Bernoullis. Under P_0 they are $\mathrm{Bern}(p_i)$; under P_1 they are $\mathrm{Bern}(p_i - \delta)$ for relevant p_i. The conditional KL divergence is:

$$\mathrm{KL}(P_0^m \| P_1^m \mid \mathcal{S}) = \sum_{i : p_i \approx 1-\varepsilon} \mathrm{KL}(\mathrm{Bern}(p_i) \| \mathrm{Bern}(p_i - \delta)) \approx \frac{m\delta^2}{\varepsilon} \tag{27}$$

identical to Theorem 7. Applying Le Cam's lemma to the conditional problem:

$$R^*_{\mathcal{S}} \geq \frac{\delta}{2} e^{-\mathrm{KL}} = c\sqrt{\frac{\varepsilon}{m}} \tag{28}$$

□

Corollary 17 (Parametric Exception)**.** *If $\mathcal{S}$ reveals that $\Delta \in \mathcal{F}_d$ for a known d-dimensional parametric family (e.g., temperature scaling: $\Delta(p) = \sigma(\mathrm{logit}(p)/T) - p$ with unknown $T \in \mathbb{R}$), then the function class shrinks from infinite-dimensional $\mathcal{F}(L)$ to a d-dimensional manifold, and verification requires only $m = O(d/\varepsilon)$ samples. The rate improves from $(L\varepsilon/m)^{1/3}$ to the parametric $O(1/\sqrt{m})$.*

White-box access helps not through the model internals per se, but by constraining the calibration function class. If the parametric form is not trusted, the nonparametric rate applies regardless of model access.

3.2 Adversarial Miscalibration Cannot Beat the Tax

A strategic developer, anticipating an audit, might shape the calibration function to minimize detectability. We show this is futile: the verification tax is the same whether miscalibration is adversarial or accidental.

Proposition 18 (Ungameability of the Verification Tax)**.** *Let a developer choose any $\Delta \in \mathcal{F}(L)$ with $\mathrm{ECE}(\Delta) = \delta$, seeking to minimize the auditor's ability to detect miscalibration. Let the auditor use any estimator on m samples. The minimax value of this verification game satisfies:*

$$\inf_{\widehat{\mathrm{ECE}}} \sup_{\substack{\Delta \in \mathcal{F}(L) \\ \mathrm{ECE}(\Delta) = \delta}} \mathbb{E}\left[|\widehat{\mathrm{ECE}} - \mathrm{ECE}(\Delta)| \right] = \Theta\left(\sqrt{\frac{\varepsilon}{m}} \right) \tag{29}$$

The developer's optimal strategy is constant miscalibration $\Delta(p) = \delta$ near $p = 1 - \varepsilon$. No sophisticated adversarial pattern improves upon this.

Proof. The lower bound is Theorem 7: the Le Cam construction already uses constant Δ, which is the hardest case. For the converse, note that any Δ with $\mathrm{ECE} = \delta$ satisfies $\|\Delta\|_1 = \delta$ (by definition) and $\|\Delta\|_2 \geq \|\Delta\|_1$ with equality iff Δ is constant on its support. The KL divergence between the calibrated and miscalibrated hypotheses is $m\|\Delta\|_2^2/\varepsilon \geq m\delta^2/\varepsilon$, with equality for constant Δ. Therefore constant miscalibration minimizes the KL, making detection hardest. Any non-constant Δ with the same ECE has strictly larger $\|\Delta\|_2$, hence larger KL, hence is *easier* to detect. □

Remark 19. This result has a direct regulatory implication: auditors need not design detection protocols for adversarial miscalibration patterns. The standard holdout-based ECE estimator is robust to strategic behavior by the model developer.

3.3 Active Verification

In passive verification, the auditor receives test samples with scores drawn from the natural distribution μ. Most samples land on "easy" confidence levels where $|\Delta(p)|$ is large and the sign is known, while the informative region near zero-crossings receives proportionally few samples. We ask: if the auditor can *choose* which confidence levels to query, does verification become cheaper?

Theorem 20 (Active Verification Rate)**.** *For the class of L-Lipschitz calibration functions with error rate ε, adaptive verification — where the auditor sequentially selects confidence levels $p_1, p_2, \ldots, p_m$, with each choice depending on all previous observations — achieves minimax rate:*

$$R^*_{\text{active}}(m, \varepsilon, L) = \Theta\left(\sqrt{\frac{\varepsilon}{m}}\right) \tag{30}$$

In particular: (i) no adaptive strategy can beat $c\sqrt{\varepsilon/m}$, and (ii) a two-phase explore-exploit strategy achieves $C\sqrt{\varepsilon/m}$, with universal constants $c, C > 0$.

Proof sketch (full proof in Appendix C). **Lower bound.** The Le Cam bound from Theorem 7 applies unchanged to adaptive strategies. Under adaptive querying, the auditor selects p_i based on all previous observations. By the chain rule of KL divergence, each term is at most δ^2/ε (the per-observation KL when p_i is in the miscalibrated region). Adaptivity does not increase the per-observation information: conditioned on p_i, the label y_i is a single Bernoulli draw whose KL is δ^2/ε regardless of previous observations. Therefore $\text{KL} \leq m\delta^2/\varepsilon$, and Le Cam gives $R^*_{\text{active}} \geq c\sqrt{\varepsilon/m}$.

Upper bound. A two-phase strategy achieves this rate: Phase 1 (exploration) queries equally-spaced confidence levels to classify bins as *resolved* (sign of Δ known) or *unresolved*; Phase 2 (exploitation) concentrates queries on resolved bins where $|\Delta|$ can be estimated without the absolute value problem. Choosing the number of grid points $N = L\sqrt{m/\varepsilon}$ balances both contributions at $O(\sqrt{\varepsilon/m})$. □

Remark 21 (Passive vs. Active Gap)**.** The gap between passive verification ($\Theta((L\varepsilon/m)^{1/3})$, Theorems 8–9) and active verification ($\Theta(\sqrt{\varepsilon/m})$, Theorem 20) is $(L^2 m/\varepsilon)^{1/6}$. For $L = 1$, $\varepsilon = 0.05$, $m = 10{,}000$, active verification is $\sim 8\times$ more sample-efficient. Crucially, the Lipschitz constant L disappears entirely from the active rate — adaptive querying eliminates the effect of calibration function smoothness.

3.4 Sequential Verification

In practice, auditors rarely pre-commit to a fixed sample budget m. Instead, they collect data until sufficiently confident, then stop. We characterize the expected stopping time of optimal sequential verification.

Definition 22 (Sequential Verification Protocol)**.** A sequential verification protocol is a pair $(\tau, \widehat{\text{ECE}}_\tau)$ where τ is a stopping time adapted to the filtration $\mathcal{F}_t = \sigma(p_1, y_1, \ldots, p_t, y_t)$ and $\widehat{\text{ECE}}_\tau$ is the ECE estimate at time τ. The protocol has accuracy δ at confidence level $1 - \alpha$ if $\Pr(|\widehat{\text{ECE}}_\tau - \text{ECE}| > \delta) \leq \alpha$.

Theorem 23 (Sequential Verification)**.** *For a model with calibration gap Δ and error rate ε:*

(i) Lower bound: *Any sequential protocol with accuracy δ at confidence $1 - \alpha$ satisfies:*

$$\mathbb{E}[\tau] \geq c \cdot \frac{\varepsilon}{\delta^2} \cdot \log \frac{1}{\alpha} \tag{31}$$

(ii) Upper bound: *There exists a sequential protocol achieving accuracy δ at confidence $1-\alpha$ with:*

$$\mathbb{E}[\tau] \leq C \cdot \frac{\varepsilon}{\delta^2} \cdot \log \frac{1}{\alpha} \tag{32}$$

The expected stopping time is $\Theta(\varepsilon/\delta^2 \cdot \log(1/\alpha))$, matching the fixed-sample Le Cam bound with an additional $\log(1/\alpha)$ factor for confidence control.

Proof sketch (full proof in Appendix D). Lower bound. Under the two Le Cam hypotheses (ECE $= 0$ vs ECE $= \delta$), the log-likelihood ratio after t observations is a random walk with drift δ^2/ε per step. By Wald's identity for sequential testing, any test with error probability $\leq \alpha$ requires expected sample size $\mathbb{E}[\tau] \geq \log(1/\alpha)/(\delta^2/\varepsilon) = \varepsilon \log(1/\alpha)/\delta^2$.

Upper bound. Construct a confidence sequence (a time-uniform confidence interval) for the ECE using the method of mixtures (Howard et al., 2021). At each time t, the confidence sequence has width $O(\sqrt{\varepsilon \log(1/\alpha)/t})$. Stop at the first τ such that the width $\leq \delta$. Solving: $\tau \approx \varepsilon \log(1/\alpha)/\delta^2$. □

Remark 24 (Instance-Dependent Behavior)**.** Sequential verification is instance-dependent. If the model is badly miscalibrated (ECE $= \delta_0 \gg \delta$), the confidence sequence excludes zero after $O(\varepsilon/\delta_0^2)$ samples — potentially very few. If the model is well-calibrated (ECE ≈ 0), confirming this to accuracy δ requires the full $O(\varepsilon/\delta^2)$ samples. Bad models are cheap to catch; good models are expensive to verify. A regulator auditing 100 models will spend most of the budget on the well-calibrated ones.

3.5 Self-Verification Impossibility

A natural question is whether external labels can be avoided entirely: can a model verify its own calibration using internal signals such as activations, dropout uncertainty, or ensemble disagreement?

Theorem 25 (Self-Verification Impossibility)**.** *Let f be a classifier with confidence scores $p(x)$ and let $z(x) = g(x, \theta)$ be any auxiliary signal derived from the model (internal activations, MC dropout variance, ensemble disagreement, attention entropy, or any other function of the input and model weights). Any estimator $V(x_1, p_1, z_1, \ldots, x_m, p_m, z_m)$ that does not use true labels y_i satisfies:*

$$\sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}\left[|V - \mathrm{ECE}(\Delta)|\right] \geq \frac{L}{2} \tag{33}$$

That is, self-verification without ground truth has worst-case error proportional to the Lipschitz constant, regardless of the number of queries m.

Proof. Fix any model with weights θ. The outputs $(p(x_i), z(x_i))$ are deterministic functions of x_i and θ; they do not depend on the calibration gap Δ. Construct two calibration functions:

- $\Delta_0(p) = 0$ for all p. The model is perfectly calibrated: $\mathrm{ECE}(\Delta_0) = 0$.
- $\Delta_1(p) = \delta$ for all p near $1-\varepsilon$, extended as an L-Lipschitz function. Then $\mathrm{ECE}(\Delta_1) = \delta$ for any $\delta \leq L$.

Both produce identical observations $(x_i, p_i, z_i)_{i=1}^m$ because the model is the same under both. Any label-free estimator V returns the same value under Δ_0 and Δ_1. Therefore:

$$\mathbb{E}[|V - \mathrm{ECE}(\Delta_0)|] + \mathbb{E}[|V - \mathrm{ECE}(\Delta_1)|] \geq |\mathrm{ECE}(\Delta_0) - \mathrm{ECE}(\Delta_1)| = \delta \tag{34}$$

Taking $\delta = L$ and the supremum: $\sup_\Delta \mathbb{E}[|V - \mathrm{ECE}|] \geq L/2$. □

Corollary 26 (Pseudo-Label Circularity)**.** *If the model generates its own labels $\hat{y}_i = \arg\max f(x_i)$ and computes ECE using $(p_i, \hat{y}_i)$, the resulting estimate equals zero for any deterministic classifier. Self-labeling measures internal consistency, not calibration.*

Corollary 27 (Distribution Shift Vulnerability)**.** *A calibration monitor trained on D_{train} (where the model was calibrated) cannot detect miscalibration on D_{test} without labeled samples from D_{test}. The monitor is a model-derived signal, falling under Theorem 25. Its worst-case error on D_{test} is at least $|\text{ECE}(D_{\text{test}}) - \text{ECE}(D_{\text{train}})|$.*

Corollary 28 (Regulatory Implication)**.** *Self-reported calibration claims by model developers have no information-theoretic value. Independent auditing with fresh labeled data is necessary, not merely desirable.*

3.6 Temperature Scaling Rate

Proposition 29 (Temperature Scaling Rate)**.** *Under correct specification (the calibration map belongs to the temperature scaling family $\Delta(p; T) = \sigma(\text{logit}(p)/T) - p$ with true parameter $T^* = 1$), the MLE $\hat{T}$ on m samples achieves:*

$$\text{CE}_{\text{smooth}} = O_p\left(\frac{1}{\sqrt{m}}\right) \tag{35}$$

for the smooth calibration error. Via the Bridge Lemma, $\text{ECE}_B = O(1/\sqrt{m} + L/B)$.

Proof sketch (full proof in Appendix E). The temperature MLE $\hat{T}$ is the unique minimizer of a 1D convex negative log-likelihood. By standard M-estimation theory (van der Vaart, 1998), $\sqrt{m}(\hat{T} - T^*) \xrightarrow{d} N(0, \mathcal{I}(T^*)^{-1})$ where $\mathcal{I}(1) = \mathbb{E}[p(1-p) \cdot \text{logit}(p)^2] \approx \varepsilon \cdot \log^2(1/\varepsilon)$. Taylor-expanding $\Delta(p; \hat{T})$ around $T^* = 1$ and integrating gives $\text{CE}_{\text{smooth}} = O_p(|\hat{T} - 1|) = O_p(1/\sqrt{m})$. Via the Bridge Lemma with $B = O(\sqrt{m}/L)$, this transfers to $\text{ECE}_B = O(1/\sqrt{m})$.

Caveat: Under misspecification, the $1/\sqrt{m}$ rate applies only to the parametric error; any misspecification bias persists regardless of m. □

Narrative role: Theorems 7–9 say verification is fundamentally expensive. Proposition 29 says: unless you assume a parametric form — then it is cheap. Theorem 16 ties them together: white-box access helps only insofar as it justifies the parametric assumption.

3.7 Further Corollaries

Corollary 30 (Fairness Extension)**.** *For a demographic group with population proportion π, verifying equalized odds requires:*

$$m \geq \frac{1}{\varepsilon \cdot \pi} \tag{36}$$

This follows from Theorem 7 applied to the subgroup with effective sample size $m \cdot \pi$.

Corollary 31 (Verification Scaling Law)**.** *If model capability scales as $\varepsilon(N) = c_0 N^{-\alpha}$ (where N is parameter count), then verification requires:*

$$m(N) = \Omega\left(N^{\alpha}\right) \tag{37}$$

samples. Verification cost grows polynomially with model size.

3.8 Verification Transfer

Proposition 32 (Verification Transfer)**.** *Let M_1 have verified calibration (ECE known to accuracy δ_1). Let M_2 be obtained from M_1 by a procedure that shifts the calibration gap by at most δ_{ft} in L^∞, with the shift function $h = \Delta_2 - \Delta_1$ being L_h-Lipschitz. To verify M_2's ECE to accuracy δ:*

(i) ***From scratch (no transfer):*** *$m_2 = \Theta(L\varepsilon_2/\delta^3)$ [from Theorem 9]*

(ii) ***With transfer:*** *$m_2 = \Theta(L_h\varepsilon_2/\delta_{\text{ft}}^3)$ suffices, provided $\delta_{\text{ft}} \leq \delta$.*

Transfer saving factor: $(L/L_h) \cdot (\delta/\delta_{\text{ft}})^3$.

Proof sketch (full proof in Appendix F). Since Δ_1 is known, the problem reduces to estimating the shift $h = \Delta_2 - \Delta_1$. A Le Cam argument with KL $\approx m_2\delta_{\text{ft}}^2/\varepsilon_2$ shows detection requires $m_2 = \Omega(\varepsilon_2/\delta_{\text{ft}}^2)$. Applying Theorem 9 to h (with Lipschitz constant L_h instead of L) and optimizing the bin count gives estimation error $O((L_h\varepsilon_2/m_2)^{1/3})$. Setting this equal to δ_{ft} yields the stated sample complexity.

Transfer helps when $\delta_{\text{ft}} < \delta$ and $L_h \leq L$. With a gentle fine-tune ($\delta_{\text{ft}} = \delta/10$), the saving is $1000\times$ fewer samples. Transfer provides no benefit when $\delta_{\text{ft}} \geq \delta$. □

4 Empirical Study

Our theoretical results generate several concrete, falsifiable predictions. We state these predictions here to facilitate independent replication.

Prediction 1: ECE scaling exponent. For a model with error rate ε, the empirical ECE estimated from m holdout samples should scale as $\widehat{\text{ECE}} \propto m^{-\alpha}$ with $\alpha \in [0.30, 0.36]$ (the theoretical prediction is $\alpha = 1/3$ from Theorem 9). Deviations toward $\alpha = 0.5$ would indicate the parametric regime (Proposition 29).

Prediction 2: Phase transition location. The transition from "insufficient data" to "meaningful verification" should occur at $m^* \approx 1/\varepsilon$ (Corollary 10). For a model with $\varepsilon = 0.05$, this predicts $m^* \approx 200$ samples.

Prediction 3: Calibration efficiency. Define calibration efficiency $\eta_{\text{eff}} = \delta_{\text{floor}}/\widehat{\text{ECE}}$, the ratio of the theoretical floor to the empirical ECE. Across models, η_{eff} should cluster below 1 (the floor is a lower bound) with η_{eff} increasing for models with smoother calibration functions.

Prediction 4: Active vs. passive gap. Under adaptive evaluation (querying high-uncertainty confidence regions preferentially), the ECE scaling exponent should shift from $\alpha \approx 1/3$ toward $\alpha \approx 1/2$ (Theorem 20).

Prediction 5: Compositional cost. For a two-stage pipeline (retriever + classifier), end-to-end ECE verification should require measurably more samples than single-model verification, with the gap scaling with the estimated system Lipschitz constant (Theorem 47).

4.1 Synthetic Validation

Why synthetic validation. Synthetic experiments are the gold standard for validating minimax rates, and are strictly more informative than experiments on real models for this purpose. Our theorems predict convergence rates as a function of three quantities: sample size m, error rate ε, and Lipschitz constant L. Validating these rates requires knowing the *true* ECE, ε, and L exactly — which is possible only when the data-generating process is fully specified. With real models, none of these quantities are known: the true ECE is itself the object of estimation, L is unobservable, and ε is estimated with noise. Experiments on real models would therefore test the theory against noisy proxies of the ground truth, introducing confounds that obscure the very rates we seek to validate. By contrast, synthetic experiments with known calibration functions provide exact ground truth, enabling sharp verification of the predicted exponents, phase transitions, and active-passive gaps. Real-world applicability is demonstrated separately through the verification audit (Table 1), which applies our formulas to published evaluation results.

5 Practical Tools

The theoretical results yield concrete formulas that practitioners can use immediately, without waiting for empirical calibration. The formulas below are parameterized by the Lipschitz

constant L; practitioners should estimate L from their evaluation data (e.g., via finite-difference approximation of the empirical calibration curve) rather than assuming $L = 1$.

Fix 1: ε-adaptive holdout sizing. For *detection* of miscalibration (testing ECE > 0 vs ECE $= 0$), the Le Cam bound (Theorem 7) gives the minimum sample size:

$$m_{\text{detect}} \geq c_1(\varepsilon)^2 \cdot \frac{\varepsilon}{\delta^2} \tag{38}$$

where $c_1(\varepsilon)$ is given in Table 3. This suffices to *detect* miscalibration of magnitude δ but does not guarantee accurate *estimation* of ECE.

For estimation of ECE to accuracy δ (the more demanding goal), Theorem 9 gives:

$$m_{\text{estimate}} \geq \frac{L\varepsilon}{\delta^3} \tag{39}$$

This is always larger than m_{detect} and is the formula practitioners should use for holdout sizing. For example, with $L = 1$, $\varepsilon = 0.05$, and $\delta = 0.02$: $m_{\text{detect}} \geq 18$ (Le Cam) but $m_{\text{estimate}} \geq 6{,}250$ (nonparametric). Using the Le Cam formula alone would undersize the holdout by over $300\times$.

We recommend practitioners use m_{estimate} as the primary sizing formula, with m_{detect} serving as a quick feasibility check (if even detection is impossible, estimation certainly is).

Fix 3: Stratified evaluation report. For multi-domain benchmarks, per-domain verification is essential because error rates vary across domains. For a domain d with n_d samples and error rate ε_d, the per-domain verification floor is $\delta_{\text{floor},d} = (L\varepsilon_d/n_d)^{1/3}$. Domains where $\delta_{\text{floor},d} > \varepsilon_d$ (i.e., $n_d < L/\varepsilon_d^2$) have insufficient data for meaningful calibration assessment and should be flagged. For MMLU's 57 subjects merged into 10–12 super-categories, this produces a table reporting: ECE estimate, verification floor, data sufficiency flag, and additional samples required per category.

Fix 4: Verification budget calculator. Given target ECE accuracy δ, confidence level $1 - \alpha$, error rates $\varepsilon_1, \ldots, \varepsilon_D$ across D domains, and group proportions $\pi_1, \ldots, \pi_G$ for G demographic groups, the minimum total holdout budget is:

$$M_{\text{total}} = \max\left\{\max_d \frac{L\varepsilon_d}{\delta^3},\ \max_g \frac{L\varepsilon}{\pi_g \delta^3}\right\} \tag{40}$$

The first term ensures per-domain verification; the second ensures per-group fairness verification. The binding constraint is typically the smallest group, consistent with the fairness verification tax (Corollary 14).

Fix 2: ε-adaptive bin count. Theorem 9 prescribes the optimal number of bins as a function of sample size, error rate, and calibration smoothness:

$$B^* = \left\lfloor \left(\frac{L^2 m}{\varepsilon}\right)^{1/3} \right\rfloor \tag{41}$$

For a model with $\varepsilon = 0.05$, $L = 1$, and $m = 10{,}000$: $B^* \approx 58$. The standard practice of using $B = 10$ or $B = 15$ bins is suboptimal for large evaluation sets and near-optimal models. This formula is justified by the matched upper bound (Theorem 9); the lower bound (Theorem 8) matches up to logarithmic factors.

Verification audit. To illustrate the practical tools, Table 1 applies the holdout sizing formula and adaptive bin count to representative models at different error rates.

6 The Verification Horizon

The verification tax has implications beyond any single audit. As models improve according to scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022), verification costs grow. Since

Table 1: Verification audit: optimal bin counts and verification floors for representative models ($L = 1$, $m = 10{,}000$). Required m uses the nonparametric rate $m \geq L\varepsilon/\delta^3$ for estimation accuracy $\delta = 0.01$.

Model class	ε	$c_1(\varepsilon)$	Required m	B^* ($m = 10$k)	Verification floor
Frontier LLM	0.05	0.347	50,000	58	0.0171
Mid-range LLM	0.15	0.321	150,000	39	0.0247
Open-source LLM	0.25	0.298	250,000	34	0.0292
Smaller model	0.35	0.276	350,000	30	0.0327

labeled data is finite, there exists a critical model size beyond which verification becomes impossible — not merely expensive, but physically infeasible given the world's supply of labeled data. We formalize this *verification horizon* and characterize the conditions under which deployment feedback can push the horizon back.

Theorem 33 (Verification Horizon)**.** *Let $\varepsilon(N) = c_0 N^{-\alpha}$ be the error rate scaling law, where N is the parameter count, $c_0 > 0$, and $\alpha > 0$. Let M_{total} be the total available labeled data. Define the* verification floor *as the best ECE accuracy achievable with M_{total} samples:*

$$\delta_{\text{floor}}(N) = \left(\frac{L\,\varepsilon(N)}{M_{\text{total}}}\right)^{1/3} = \left(\frac{Lc_0}{M_{\text{total}}}\right)^{1/3} N^{-\alpha/3} \tag{42}$$

Verification is meaningful *only when $\delta_{\text{floor}}(N) < \varepsilon(N)$, i.e., when the verification accuracy is finer than the quantity being measured. This holds if and only if $N < N^*$, where:*

$$N^* = \left(\frac{c_0^2 M_{\text{total}}}{L}\right)^{1/(2\alpha)} \tag{43}$$

For $N > N^*$, no verifier — regardless of algorithm, adaptivity, or computational resources — can verify calibration to a precision finer than the model's actual ECE. The model enters a regime of unverifiable improvement*: it may be getting better, but nobody can confirm this.*

Proof. From Theorem 9, verification to accuracy δ requires $m \geq L\varepsilon/\delta^3$. With $m = M_{\text{total}}$, the best achievable accuracy is $\delta_{\text{floor}} = (L\varepsilon/M_{\text{total}})^{1/3}$. Verification is meaningful when $\delta_{\text{floor}} < \varepsilon$:

$$\left(\frac{L\varepsilon}{M_{\text{total}}}\right)^{1/3} < \varepsilon \implies \frac{L}{M_{\text{total}}} < \varepsilon^2 \implies \varepsilon > \sqrt{\frac{L}{M_{\text{total}}}} \tag{44}$$

Substituting $\varepsilon(N) = c_0 N^{-\alpha}$:

$$c_0 N^{-\alpha} > \sqrt{\frac{L}{M_{\text{total}}}} \implies N < \left(\frac{c_0^2 M_{\text{total}}}{L}\right)^{1/(2\alpha)} = N^* \tag{45}$$

□

Remark 34 (Quantitative Estimates)**.** Under Chinchilla-type scaling ($\alpha \approx 0.5$, $c_0 \approx 1$) with $L = 1$ and $M_{\text{total}} = 10^7$ (approximate total expert-labeled evaluation data across major benchmarks): $N^* = (10^7)^{1/1} = 10^7$. Current frontier models exceed 10^{11} parameters — well past this naive estimate. However, M_{total} grows with benchmark creation, and active verification (Theorem 20) shifts the horizon. Under active verification, the condition becomes $\varepsilon > 1/M_{\text{total}}$ (from the $\sqrt{\varepsilon/m}$ rate), giving $N^*_{\text{active}} = (c_0 M_{\text{total}})^{1/\alpha}$, which is strictly larger. Adaptive evaluation buys time but does not eliminate the horizon.

Corollary 35 (Active Verification Horizon)**.** *Under active verification (Theorem 20), the verification floor improves to $\delta_{\text{floor,active}} = \sqrt{\varepsilon/M_{\text{total}}}$, and the active horizon is:*

$$N^*_{\text{active}} = (c_0 M_{\text{total}})^{1/\alpha} \tag{46}$$

Note that L does not appear: the active rate $\Theta(\sqrt{\varepsilon/m})$ is Lipschitz-free. The ratio $N^*_{\text{active}}/N^* = (c_0 M_{\text{total}})^{1/\alpha}/(c_0^2 M_{\text{total}}/L)^{1/(2\alpha)}$ quantifies how much additional runway adaptive evaluation provides.

Theorem 36 (Verification Dynamics)**.** *Suppose a model is deployed at rate r samples per unit time, and each deployment yields a verified label (e.g., through delayed ground truth). At time t, the cumulative labeled data is $m(t) = rt$, and the model's error rate (potentially improving through retraining) is $\varepsilon(t) = c_0 t^{-\beta}$ for some improvement exponent $\beta > 0$.*

(i) If $\beta < 1/2$: the verification floor $\delta_{\text{floor}}(t) = (L\varepsilon(t)/rt)^{1/3}$ eventually falls below $\varepsilon(t)$. Verification catches up*: there exists t_0 such that for all $t > t_0$, meaningful verification is possible.*

(ii) If $\beta > 1/2$: the verification floor eventually exceeds $\varepsilon(t)$. Verification falls behind*: there exists t_1 such that for all $t > t_1$, meaningful verification is impossible despite growing data.*

(iii) If $\beta = 1/2$: verification is perpetually marginal. Meaningful verification is possible if and only if $L/(rc_0^2) < 1$.

The critical improvement exponent is $\beta^* = 1/2$.

Proof. The verification floor at time t is:

$$\delta_{\text{floor}}(t) = \left(\frac{L\varepsilon(t)}{rt}\right)^{1/3} = \left(\frac{Lc_0}{r}\right)^{1/3} t^{-(\beta+1)/3} \tag{47}$$

Verification is meaningful when $\delta_{\text{floor}}(t) < \varepsilon(t) = c_0 t^{-\beta}$:

$$\left(\frac{Lc_0}{r}\right)^{1/3} t^{-(\beta+1)/3} < c_0 t^{-\beta} \tag{48}$$

Simplifying:

$$\left(\frac{L}{rc_0^2}\right)^{1/3} < t^{-\beta+(\beta+1)/3} = t^{(1-2\beta)/3} \tag{49}$$

If $\beta < 1/2$: the exponent $(1 - 2\beta)/3 > 0$, so the RHS grows with t. The inequality is satisfied for t large enough.

If $\beta > 1/2$: the exponent $(1 - 2\beta)/3 < 0$, so the RHS shrinks to zero. The inequality is eventually violated.

If $\beta = 1/2$: the exponent is zero. The inequality reduces to $L/(rc_0^2) < 1$, which is a fixed condition independent of t. □

Remark 37 (Interpretation)**.** The critical exponent $\beta^* = 1/2$ has a clean interpretation. Verification cost grows as ε^{-2} (from the meaningful verification condition $m > 1/\varepsilon^2$). Data grows linearly in time. If the model improves so fast that ε^{-2} grows faster than t — i.e., ε shrinks faster than $t^{-1/2}$ — verification loses the race. This is independent of the Lipschitz constant, the deployment rate, and the verification algorithm. It is a fundamental speed limit on how fast models can improve while remaining verifiable.

Remark 38 (Active Verification Dynamics)**.** Under active verification (Theorem 20), the verification floor is $\delta_{\text{floor,active}}(t) = \sqrt{\varepsilon(t)/rt}$, and the meaningful verification condition becomes $\varepsilon(t) > 1/(rt)$, i.e., $t^{1-\beta} > 1/(rc_0)$. The critical exponent shifts to $\beta^*_{\text{active}} = 1$. Adaptive evaluation doubles the critical exponent — models can improve twice as fast (in the scaling exponent) while remaining actively verifiable. But the horizon still exists.

6.1 The Recalibration Trap

Iterative recalibration — via temperature scaling, Platt scaling, or RLHF — is the standard remedy for miscalibrated models. Each round reduces the ECE. But each round also makes the *next* improvement harder to verify, because the remaining miscalibration is smaller and requires more data to measure. We show that the number of verifiable recalibration rounds is sharply bounded.

Proposition 39 (Recalibration Trap)**.** *Consider iterative recalibration that reduces ECE by a factor $\gamma \in (0,1)$ per round, so that $\mathrm{ECE}_K = \gamma^K \cdot \mathrm{ECE}_0$ after K rounds. The verification cost of confirming that round $K+1$ produced a genuine improvement is:*

$$m_K = \Omega\left(\frac{\varepsilon}{\gamma^{2K} \cdot \mathrm{ECE}_0^2 \cdot (1-\gamma)^2}\right) \tag{50}$$

which grows as $(1/\gamma^2)^K$ — exponential in the number of rounds. Given a fixed evaluation budget M_{total}, the maximum number of verifiable rounds is:

$$K^* = \left\lfloor \frac{\log\left(M_{\text{total}} \cdot (1-\gamma)^2 \cdot \mathrm{ECE}_0^2/\varepsilon\right)}{2\log(1/\gamma)} \right\rfloor \tag{51}$$

Beyond round K^*, continued recalibration is unverifiable: the model may be improving, but the evaluation data cannot confirm it.

Proof. At round K, the ECE is $\mathrm{ECE}_K = \gamma^K \mathrm{ECE}_0$. The improvement from round K to $K+1$ is:

$$\Delta_K = \mathrm{ECE}_K - \mathrm{ECE}_{K+1} = \gamma^K \mathrm{ECE}_0 (1-\gamma) \tag{52}$$

By Theorem 7, distinguishing ECE_K from ECE_{K+1} requires:

$$m_K \geq \frac{\varepsilon}{\Delta_K^2} = \frac{\varepsilon}{\gamma^{2K} \mathrm{ECE}_0^2 (1-\gamma)^2} \tag{53}$$

Setting $m_K = M_{\text{total}}$ and solving for K gives K^*. □

Remark 40 (Worked Example)**.** Consider $\gamma = 0.5$ (each round halves ECE), $\varepsilon = 0.05$, $\mathrm{ECE}_0 = 0.10$, and $M_{\text{total}} = 14{,}000$ (MMLU). Then:

$$K^* = \left\lfloor \frac{\log(14000 \cdot 0.25 \cdot 0.01/0.05)}{2\log 2} \right\rfloor = \left\lfloor \frac{\log(700)}{1.386} \right\rfloor = \left\lfloor \frac{6.55}{1.386} \right\rfloor = 4 \tag{54}$$

After 4 rounds of recalibration, MMLU cannot verify further improvement. This is not a limitation of the evaluation methodology — it is a fundamental information-theoretic barrier. Every organization performing iterative RLHF calibration tuning faces this ceiling.

Remark 41. The recalibration trap implies that regulatory requirements for "continuous calibration improvement" are infeasible beyond K^* rounds without proportionally growing the evaluation budget.

6.2 Verification Half-Life

Every verification result has an expiry date. A model verified today may become miscalibrated tomorrow as the deployment distribution shifts — user demographics change, language evolves, disease prevalence fluctuates, market regimes shift. We formalize this temporal decay of verification and derive the minimum data collection rate required to maintain perpetual assurance.

Definition 42 (ECE Drift Rate)**.** The *ECE drift rate* $\lambda > 0$ bounds the rate at which the true ECE changes over time:

$$|\mathrm{ECE}(t) - \mathrm{ECE}(0)| \leq \lambda \cdot t \tag{55}$$

for all $t \geq 0$. This is implied by a distribution drift rate ρ (total variation distance per unit time) and an L-Lipschitz calibration function: $\lambda \leq L \cdot \rho$.

Theorem 43 (Verification Half-Life)**.** *Let a model be verified at time $t = 0$ with ECE accuracy δ (i.e., $|\widehat{\mathrm{ECE}} - \mathrm{ECE}(0)| \leq \delta$), and let the ECE drift rate be λ.*

(i) Half-life: *The verification becomes invalid (accuracy exceeds 2δ) after:*

$$t_{1/2} = \frac{\delta}{\lambda} \tag{56}$$

(ii) Perpetual verification: *Maintaining ECE accuracy δ at all times, by re-verifying every $t_{1/2}$ time units, requires a sustained data collection rate of:*

$$r \geq \frac{L\varepsilon\lambda}{\delta^4} \tag{57}$$

labeled samples per unit time. Below this rate, verification permanently expires.

(iii) Accelerating drift: *If $\lambda(t)$ is increasing, the half-life $t_{1/2}(t) = \delta/\lambda(t) \to 0$ and the required data rate $r(t) \to \infty$. Eventually, no data collection rate suffices.*

Proof. Part (i). At time t, the triangle inequality gives:

$$|\widehat{\mathrm{ECE}} - \mathrm{ECE}(t)| \leq \underbrace{|\widehat{\mathrm{ECE}} - \mathrm{ECE}(0)|}_{\leq\delta} + \underbrace{|\mathrm{ECE}(0) - \mathrm{ECE}(t)|}_{\leq\lambda t} \leq \delta + \lambda t \tag{58}$$

This exceeds 2δ when $\lambda t > \delta$, giving $t_{1/2} = \delta/\lambda$.

Part (ii). Re-verification every $t_{1/2}$ time units costs $m = L\varepsilon/\delta^3$ samples each (from Theorem 9). Data needed per unit time: $r = m/t_{1/2} = (L\varepsilon/\delta^3) \cdot (\lambda/\delta) = L\varepsilon\lambda/\delta^4$.

Part (iii). Immediate from $t_{1/2}(t) = \delta/\lambda(t) \to 0$ and $r(t) = L\varepsilon\lambda(t)/\delta^4 \to \infty$. □

Remark 44 (Domain-Specific Half-Lives)**.** We estimate verification half-lives for four deployment domains, using representative drift rates from the literature (Davis et al., 2020):

Domain	λ	δ	$t_{1/2}$	r (samples/month)
Medical AI	0.01/month	0.02	2 months	∼3,100
Content moderation	0.005/month	0.03	6 months	∼620
Financial trading	0.1/week	0.01	17 hours	$\sim 10^6$/week
Autonomous driving	0.02/month	0.01	15 days	∼100,000

These estimates assume $L = 1$ and $\varepsilon = 0.05$. Medical AI and content moderation can maintain perpetual verification with modest data budgets. Financial models face verification half-lives so short that continuous verification is nearly infeasible. Every deployed AI system has a verification half-life; we argue it should be computed and disclosed as part of the model card.

Remark 45 (Benchmark Half-Lives)**.** Static benchmarks also have verification half-lives. A benchmark constructed at time $t = 0$ reflects the distribution D_0. As the world drifts (new topics emerge, language evolves, knowledge changes), the benchmark distribution diverges from deployment reality. Any calibration verification performed on a static benchmark decays at rate λ, regardless of the model's actual calibration on current data. This provides a formal argument for periodic benchmark refresh, complementing empirical observations of benchmark saturation and contamination.

7 Compositional Verification

Modern AI systems are rarely single models. Retrieval-augmented generation (Lewis et al., 2020), multi-step reasoning (Wei et al., 2022), and agentic pipelines (Yao et al., 2023) compose multiple components into end-to-end systems. Even if every component is individually verified as well-calibrated, the *composed system* may not be. We show that the verification cost of composed systems grows exponentially with pipeline depth — a fundamental barrier to end-to-end verification of complex AI systems.

Definition 46 (Compositional AI Pipeline)**.** A K-component AI pipeline produces a final prediction with confidence p_K through a chain of confidence transformations:

$$p_k = g_k(p_{k-1}, x), \quad k = 1, \ldots, K \tag{59}$$

where p_0 is the input confidence (or prior), x is the input, and each g_k is L_k-Lipschitz in its first argument (holding x fixed). The end-to-end calibration gap is $\Delta_{\mathrm{sys}}(p) = \eta_{\mathrm{sys}}(p) - p$ where $\eta_{\mathrm{sys}}(p) = \mathbb{E}[Y \mid p_K = p]$.

Theorem 47 (Compositional Verification Tax)**.** *Let a K-component pipeline have component confidence transformations $g_1, \ldots, g_K$ with Lipschitz constants $L_1, \ldots, L_K$. The end-to-end calibration function has Lipschitz constant:*

$$L_{\text{sys}} \leq \prod_{k=1}^{K} L_k + 1 \tag{60}$$

and end-to-end verification to accuracy δ requires:

$$m_{\text{sys}} \geq c \cdot \frac{L_{\text{sys}} \cdot \varepsilon}{\delta^3} \tag{61}$$

For homogeneous pipelines with $L_k = L > 1$ for all k:

$$m_{\text{sys}} = \Omega\left(\frac{L^K \cdot \varepsilon}{\delta^3}\right) \tag{62}$$

Verification cost grows exponentially with pipeline depth K.

Proof. Lipschitz composition. By the chain rule for Lipschitz functions, if g_k is L_k-Lipschitz in its first argument, then the composition $g_K \circ \cdots \circ g_1$ is $(\prod_{k=1}^{K} L_k)$-Lipschitz. Since $\Delta_{\text{sys}}(p) = \eta_{\text{sys}}(p) - p$ and $p \mapsto p$ is 1-Lipschitz, the triangle inequality gives:

$$L_{\text{sys}} \leq \prod_{k=1}^{K} L_k + 1 \tag{63}$$

Verification cost. Applying Theorem 9 with $L = L_{\text{sys}}$:

$$m_{\text{sys}} = \Theta\left(\frac{L_{\text{sys}} \cdot \varepsilon}{\delta^3}\right) \tag{64}$$

For homogeneous $L_k = L$: $L_{\text{sys}} \approx L^K$ (the $+1$ is negligible for $L > 1$), giving exponential scaling in K. □

Corollary 48 (Maximum Verifiable Pipeline Depth)**.** *For a fixed verification budget M_{total} and target accuracy δ, the maximum verifiable pipeline depth is:*

$$K_{\max} = \left\lfloor \log_L \frac{M_{\text{total}} \cdot \delta^3}{\varepsilon} \right\rfloor \tag{65}$$

Deeper pipelines require exponentially more data to verify.

Corollary 49 (Agent Loop Verification Bound)**.** *An AI agent executing K reasoning iterations before producing a final answer has verification cost $\Omega(L^K \cdot \varepsilon/\delta^3)$. For $L = 2$ and $K = 10$: the verification cost is 1,024× that of a single-step model. For $K = 20$: over $10^6\times$.*

Remark 50 (Compositionality–Verifiability Tradeoff)**.** Systems designed to "think more carefully" via longer reasoning chains become exponentially harder to verify. The reliability improvement from additional reasoning must be weighed against the exponentially growing cost of confirming that improvement. This tradeoff applies to chain-of-thought prompting, iterative refinement, multi-hop retrieval, and agentic tool-use loops.

Remark 51 (On the Lipschitz Assumption)**.** The Lipschitz composition model is an optimistic assumption. In real pipelines, confidence transformations may be *discontinuous* — e.g., a retriever returning a different document set for marginally different queries. Discontinuity corresponds to $L_k = \infty$, making $L_{\text{sys}} = \infty$ and end-to-end verification impossible at *any* finite sample size. The Lipschitz model provides a lower bound on verification cost; the true cost may be higher.

As a concrete illustration, consider a two-stage retriever–classifier pipeline evaluated on MMLU. The retriever assigns a relevance confidence p_1 to retrieved context; the classifier uses this context to produce a final answer confidence p_2. If both components have Lipschitz constant $L = 2$, then $L_{\text{sys}} \leq 5$ and verification requires 5× the data of a single classifier. With $L = 3$ (a modest sensitivity), the cost rises to 10×. A three-stage pipeline with $L = 2$ requires 9× the data; with $L = 3$, 28×. These factors compound: adding a single pipeline stage at $L = 2$ always doubles the Lipschitz-dependent verification cost.

Table 2: Comparison with prior work on calibration estimation.

Paper	Their Result	Our Differentiation
Sun et al. (2023)	UB $O(n^{-2/3})$, $B \sim n^{1/3}$	Matching LB + ε-dependence
Futami & Fujisawa (2024)	General $n^{-1/3}$ UB for ECE bias	We add ε-dependence + first LB
Futami & Nitanda (2026)	Smooth CE uniform convergence	Different question: data requirements
Hu et al. (2024)	$\Omega(\varepsilon^{-2.5})$ testing complexity	Testing vs. estimation; different ε
Lepski et al. (1999)	L_1 norm minimax rate	We specialize to calibration with ε noise
Blasiok et al. (2023)	Calibration distance framework	Different metric (distance vs. ECE)
Kossen et al. (2021)	Active testing framework	We derive minimax-optimal active rate; they provide algor

8 Related Work

Calibration estimation. The statistical properties of ECE estimation have been studied by several recent works. Sun, Song & Hero (Sun et al., 2023) established an upper bound of $O(n^{-2/3})$ for histogram binning with $B \sim n^{1/3}$ bins and proposed a minimum-risk recalibration method. Futami & Fujisawa (Futami & Fujisawa, 2024) provided the first comprehensive bias analysis for both uniform-width and uniform-mass binning, establishing convergence rates and optimal bin counts. Their $n^{-1/3}$ rate is an upper bound on the estimation bias; we complement it with a matching lower bound (up to logarithmic factors) and reveal the previously hidden ε-dependence. Ciosek et al. (Ciosek et al., 2025) derived finite-sample bounds under Lipschitz and bounded-variation assumptions but did not study ε-dependence or scaling laws. Our work differs from all of these in establishing both upper and lower bounds that explicitly depend on the error rate, revealing the verification tax phenomenon.

Calibration testing and distance. Hu et al. (Hu et al., 2024) studied the testing complexity of calibration distance, establishing an $\Omega(\varepsilon^{-2.5})$ sample complexity. Their problem is distinct from ours: they test whether a model is calibrated (hypothesis testing), while we estimate how miscalibrated it is (estimation). The calibration distance framework of Blasiok et al. (Blasiok et al., 2023) provides a foundational metric; our ECE-based analysis uses a different but related measure. Futami & Nitanda (Futami & Nitanda, 2026) studied uniform convergence of the smooth calibration error; their focus is on algorithm guarantees rather than data requirements.

Active evaluation. Kossen et al. (Kossen et al., 2021) introduced the active testing framework for sample-efficient model evaluation, proposing practical acquisition strategies. Our Theorem 20 provides the minimax-optimal rate for active calibration verification, showing that adaptive evaluation achieves $\Theta(\sqrt{\varepsilon/m})$ — a tight characterization that was not established in prior work.

Nonparametric functional estimation. Our lower bound technique (Theorem 8) builds on the seminal work of Lepski, Nemirovski & Spokoiny (Lepski et al., 1999), who established minimax rates for L_r norm estimation in white noise models. We identify ECE estimation in the rare-error regime as an instance of their framework, with the novel observation that the noise variance equals the error rate ε, which introduces the ε-dependence absent from their general theory. The Brown–Low asymptotic equivalence (Brown & Low, 1996) between nonparametric regression and white noise underpins the reduction.

Davis et al. (Davis et al., 2020) develop empirical methods for detecting calibration drift in clinical prediction models; our verification half-life (Theorem 43) provides the theoretical foundation, deriving the exact shelf-life of a calibration audit and the minimum data rate for perpetual assurance.

9 Conclusion

We have established a comprehensive mathematical framework for AI verification in the rare-error regime. The central result — the verification tax — states that estimating the calibration error of a model with error rate ε requires $\Theta((L\varepsilon/m)^{1/3})$ labeled samples, a cost that grows as models improve. This tax is universal across calibration, fairness, and robustness verification (Theorem 13), cannot be circumvented by model access (Theorem 16), cannot be gamed by strategic miscalibration (Proposition 18), and cannot be avoided by self-verification (Theorem 25).

Two avenues reduce the tax: parametric assumptions compress the calibration function class, achieving the faster $O(1/\sqrt{m})$ rate (Proposition 29), and adaptive evaluation protocols eliminate the dependence on calibration smoothness entirely, achieving the tight rate $\Theta(\sqrt{\varepsilon/m})$ (Theorem 20). Prior verification transfers across model updates proportionally to the magnitude of the calibration shift (Proposition 32).

At scale, the verification tax creates a verification horizon — a critical model size beyond which the world's supply of labeled data is insufficient to meaningfully verify performance (Theorem 33). Iterative recalibration hits a ceiling of $K^* = O(\log M_{\text{total}})$ verifiable rounds (Proposition 39), and compositional AI systems face verification costs that grow exponentially with pipeline depth (Theorem 47). Every verification has a half-life determined by the deployment distribution's drift rate (Theorem 43).

Limitations. Our analysis relies on the Lipschitz assumption for the calibration function, which may not hold for all models. The lower bound (Theorem 8) inherits a logarithmic gap from the open problem of L_1 functional estimation (Lepski et al., 1999). The compositional verification tax (Theorem 47) uses a Lipschitz composition model that may overestimate the effective system Lipschitz constant for pipelines with discrete components. Sharp constants (Table 3) are computed for the Le Cam bound only; constants for the minimax rate remain open.

Open problems. Three directions warrant further investigation. First, closing the logarithmic gap between the upper and lower bounds for ECE estimation — a special case of the long-standing open problem for L_1 functional estimation. Second, extending the compositional verification tax to pipelines with non-Lipschitz components (e.g., discrete retrieval steps). Third, characterizing the verification tax for multi-class calibration, where the calibration gap becomes a vector-valued function.

Broader implications. The verification tax has direct consequences for AI governance. Self-reported calibration claims carry no information-theoretic weight (Theorem 25). Independent auditing with fresh labeled data is necessary, not merely desirable. Regulators should invest in adaptive evaluation protocols and adopt continuous auditing frameworks with re-audit schedules determined by the verification half-life. The trend toward deeper, iteratively refined, multi-component AI systems is on a collision course with our ability to verify them.

Acknowledgments

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A Full Proof of Theorem 8

Step 1: Reduction to nonparametric regression.

Within a window of width w around $p = 1-\varepsilon$, where the score distribution concentrates, the calibration verification problem reduces to estimating the L_1 norm of a regression function. Specifically:

For $p \in [1-\varepsilon-w/2,\, 1-\varepsilon+w/2]$, define $\Delta(p) = \eta(p) - p$. The labels Y_i given $p(X_i) = p$ are $\mathrm{Bern}(\eta(p)) = \mathrm{Bern}(p + \Delta(p))$, with variance $\eta(p)(1-\eta(p)) \approx \varepsilon$ in the rare-error regime.

The ECE restricted to this window is:

$$\mathrm{ECE}_{\mathrm{window}} = \int_{1-\varepsilon-w/2}^{1-\varepsilon+w/2} |\Delta(p)|\, d\mu(p) \tag{66}$$

This is precisely the L_1 norm of Δ on the window, weighted by μ.

Step 2: Identification with L_1 functional estimation in white noise.

By the asymptotic equivalence of nonparametric regression and Gaussian white noise (Brown & Low, 1996; Nussbaum, 1996), estimating $\int |\Delta(p)| dp$ from m observations with per-observation variance ε is asymptotically equivalent to estimating $\int |f(t)| dt$ in the white noise model:

$$dX(t) = f(t)dt + (m/\varepsilon)^{-1/2} dW(t), \quad t \in [0,1] \tag{67}$$

with effective sample size $n_{\mathrm{eff}} = m/\varepsilon$ and $f \in \Sigma(\beta = 1, L)$ (Lipschitz = Hölder with $\beta = 1$).

Step 3: Application of Lepski, Nemirovski & Spokoiny (1999).

By Theorem 2.2 of Lepski et al. (1999), the minimax rate for estimating the L_1 norm of a β-Hölder function in white noise with sample size n is:

$$R^*_{L_1} \geq c \cdot n^{-\beta/(2\beta+1)} / (\log n)^{c'} \tag{68}$$

For $\beta = 1$ (Lipschitz) and $n = n_{\mathrm{eff}} = m/\varepsilon$:

$$R^*_{L_1} \geq c \cdot (m/\varepsilon)^{-1/3} / (\log(m/\varepsilon))^{c'} = c \cdot \left(\frac{\varepsilon}{m}\right)^{1/3} / (\log m)^{c'} \tag{69}$$

Incorporating the Lipschitz constant L (which scales the function class):

$$R^*(m, \varepsilon, L) \geq c_2 \left(\frac{L\varepsilon}{m}\right)^{1/3} / (\log m)^{c_3} \tag{70}$$

Technical conditions:

- *Score-mass concentration:* The score distribution μ must place mass $\geq \rho$ in a window of width w around $1-\varepsilon$. We state this as an assumption. This holds for well-trained classifiers: overconfident models concentrate predictions near the top.
- *The Brown–Low equivalence* requires m sufficiently large relative to the smoothness. The minimum m is $m \geq C(L)/\varepsilon$ for a constant depending on L.
- *Lipschitz constant:* Since $\Delta(p) = \eta(p) - p$ and $p \mapsto p$ is 1-Lipschitz, we have $L_\Delta \leq L_\eta + 1$. We use L throughout for L_Δ.

B Supplementary Details for Theorem 16

The proof of Theorem 16 in the main text is complete. We note that the result extends straightforwardly to any side information $\mathcal{S}$ that is independent of the fresh evaluation labels conditional on the scores. This includes: model weights, architecture specifications, training data, training logs, gradient histories, and any derived quantity (e.g., Fisher information matrices, Hessian spectra, pruning masks). The key insight is that calibration is a property of the model-world interface, not of the model alone.

C Full Proof of Theorem 20 (Active Verification)

Lower bound. The Le Cam lower bound from Theorem 7 applies unchanged to adaptive strategies. Construct the same two hypotheses: P_0 (calibrated, ECE $= 0$) and P_1 (miscalibrated by δ near $p = 1-\varepsilon$, ECE $= \delta$). Under adaptive querying, the auditor selects p_i based on all previous observations $(p_1, y_1, \ldots, p_{i-1}, y_{i-1})$. By the chain rule of KL divergence:

$$\mathrm{KL}(P_0^m \| P_1^m) = \sum_{i=1}^{m} \mathbb{E}\left[\mathrm{KL}(P_0(\cdot \mid p_i) \| P_1(\cdot \mid p_i))\right] \tag{71}$$

Each term is at most δ^2/ε (the per-observation KL when p_i is in the miscalibrated region, zero otherwise). Adaptivity does not increase the per-observation information: conditioned on p_i, the label y_i is a single Bernoulli draw whose KL is δ^2/ε regardless of previous observations. Therefore KL $\leq m\delta^2/\varepsilon$, and Le Cam gives $R^*_{\text{active}} \geq c\sqrt{\varepsilon/m}$.

Upper bound. We construct a two-phase adaptive strategy.

Phase 1 (Exploration): Select N equally-spaced confidence levels $\{p_j = j/N\}_{j=1}^{N}$ in the interval $[1-\varepsilon-w, 1-\varepsilon+w]$ for a window width $w = O(1)$. Query each level $m/(2N)$ times. The empirical estimate $\hat{\Delta}(p_j)$ has standard error $\sigma_j = \sqrt{2\varepsilon N/m}$. Classify each bin as:

- *Resolved*: $|\hat{\Delta}(p_j)| > 2\sigma_j$ — the sign of $\Delta(p_j)$ is known with high probability.
- *Unresolved*: $|\hat{\Delta}(p_j)| \leq 2\sigma_j$ — the sign is uncertain.

The unresolved zone has width at most $4\sigma_j/L = O(\sqrt{\varepsilon N/m}/L)$ by the Lipschitz constraint (the function can only stay near zero for a region of width $\leq |\Delta|_{\max}/L$).

Phase 2 (Exploitation): Use the remaining $m/2$ queries on resolved bins to estimate $|\Delta(p_j)| = \Delta(p_j) \cdot \mathrm{sign}(\Delta(p_j))$ (sign is known, so no absolute value problem). This is a simple average with error $\sqrt{\varepsilon/m}$.

Total error:

- Resolved bins: estimation error $O(\sqrt{\varepsilon/m})$.
- Unresolved bins: the contribution to ECE is at most (max $|\Delta|$ in zone) $\times$ (width of zone). Since $|\Delta| \leq 2\sigma_j$ in the zone and the width is $O(\sigma_j/L)$, this contributes $O(\sigma_j^2/L) = O(\varepsilon N/(mL))$.

Choose $N = L\sqrt{m/\varepsilon}$ to balance the two terms. Both become $O(\sqrt{\varepsilon/m})$. □

D Full Proof of Theorem 23 (Sequential Verification)

Lower bound. Under the two Le Cam hypotheses (P_0: ECE $= 0$, P_1: ECE $= \delta$), the log-likelihood ratio after t observations is:

$$\Lambda_t = \sum_{i=1}^{t} \log \frac{p_1(y_i \mid p_i)}{p_0(y_i \mid p_i)} \tag{72}$$

This is a random walk with per-step drift:

$$\mathbb{E}_{P_0}[\Lambda_1] = -\mathrm{KL}(P_0(\cdot \mid p_i) \| P_1(\cdot \mid p_i)) \approx -\frac{\delta^2}{\varepsilon} \tag{73}$$

By Wald's identity for sequential testing (Tsybakov, 2009), any test between P_0 and P_1 with error probabilities $\leq \alpha$ satisfies:

$$\mathbb{E}[\tau] \geq \frac{\log(1/\alpha)}{\mathrm{KL}(P_0 \| P_1)} = \frac{\varepsilon \log(1/\alpha)}{\delta^2} \tag{74}$$

Since any sequential verification protocol with accuracy δ at confidence $1-\alpha$ must distinguish $\mathrm{ECE} = 0$ from $\mathrm{ECE} = \delta$ with probability $\geq 1 - \alpha$, the lower bound follows.

Upper bound. We construct a confidence sequence for the ECE using the method of mixtures (Howard et al., 2021). Define the running ECE estimate:

$$\widehat{\mathrm{ECE}}_t = \sum_{b=1}^{B} \frac{n_{b,t}}{t} \left|\hat{\Delta}_{b,t}\right| \tag{75}$$

where $\hat{\Delta}_{b,t}$ is the empirical calibration gap in bin b after t observations.

By Theorem 1 of Howard et al. (2021), there exists a time-uniform confidence sequence $(C_t)_{t\geq 1}$ such that:

$$\Pr\left(\exists t \geq 1 : |\widehat{\mathrm{ECE}}_t - \mathrm{ECE}| > C_t\right) \leq \alpha \tag{76}$$

with $C_t = O(\sqrt{\varepsilon \log(\log(t)/\alpha)/t})$.

Define the stopping time $\tau = \inf\{t : C_t \leq \delta\}$. Solving $C_t \leq \delta$:

$$\sqrt{\frac{\varepsilon \log(\log(t)/\alpha)}{t}} \leq \delta \implies t \geq \frac{\varepsilon \log(\log(t)/\alpha)}{\delta^2} \tag{77}$$

For the dominant term, $\mathbb{E}[\tau] = O(\varepsilon \log(1/\alpha)/\delta^2)$, matching the lower bound up to constants. □

E Temperature Scaling Analysis

We provide the full proof of Proposition 29.

Step 1: MLE for the temperature parameter. Given scores $p_1, \ldots, p_m$ and labels $y_1, \ldots, y_m$, the negative log-likelihood is:

$$\ell(T) = -\sum_{i=1}^{m} \left[y_i \log \sigma(\mathrm{logit}(p_i)/T) + (1 - y_i) \log(1 - \sigma(\mathrm{logit}(p_i)/T))\right] \tag{78}$$

This is a 1D convex optimization. The MLE $\hat{T}$ is unique.

Step 2: Asymptotic normality of MLE. By standard M-estimation theory (van der Vaart, 1998):

$$\sqrt{m}(\hat{T} - T^*) \xrightarrow{d} N(0, \mathcal{I}(T^*)^{-1}) \tag{79}$$

where $\mathcal{I}(T^*)$ is the Fisher information. At $T^* = 1$, the score function is:

$$\frac{\partial}{\partial T} \log p(Y \mid p, T)\bigg|_{T=1} = -(Y - p) \cdot \mathrm{logit}(p) \tag{80}$$

So:

$$\mathcal{I}(1) = \mathbb{E}[(Y - p)^2 \cdot \mathrm{logit}(p)^2] = \mathbb{E}[p(1 - p) \cdot \mathrm{logit}(p)^2] \tag{81}$$

In the rare-error regime with scores concentrated near $1 - \varepsilon$: $\mathrm{logit}(1 - \varepsilon) \approx \log(1/\varepsilon)$ and $p(1 - p) \approx \varepsilon$, giving:

$$\mathcal{I}(1) \approx \varepsilon \cdot \log^2(1/\varepsilon) \tag{82}$$

This is positive and grows as $\varepsilon \to 0$ — rare errors make temperature estimation *easier* because logits are more spread out. The MLE satisfies: $|\hat{T} - T^*| = O_p(1/\sqrt{m \cdot \mathcal{I}(T^*)})$.

Step 3: Translating to calibration error. By Taylor expansion of $\Delta(p; T)$ around $T^* = 1$:

$$\Delta(p; \hat{T}) \approx \frac{\partial \Delta}{\partial T}\bigg|_{T=1} \cdot (\hat{T} - 1) = p(1 - p) \cdot \mathrm{logit}(p) \cdot (\hat{T} - 1) \tag{83}$$

So the smooth CE is:

$$\mathrm{CE}_{\mathrm{smooth}} = \int |\Delta(p;\hat{T})|\, d\mu(p) \approx |\hat{T}-1| \cdot \underbrace{\int p(1-p)|\mathrm{logit}(p)|\, d\mu(p)}_{\kappa} \tag{84}$$

where κ is a constant depending on the score distribution. Substituting $|\hat{T}-1| = O_p(1/\sqrt{m \cdot \mathcal{I}(T^*)})$:

$$\mathrm{CE}_{\mathrm{smooth}} = O_p\left(\frac{\kappa}{\sqrt{m \cdot \mathcal{I}(T^*)}}\right) = O_p\left(\frac{1}{\sqrt{m}}\right) \tag{85}$$

with a constant that depends on ε through $\mathcal{I}(T^*)$ and κ. This is the **parametric rate** $m^{-1/2}$, not the nonparametric rate $(L\varepsilon/m)^{1/3}$.

Step 4: Connection to binned ECE. Via the Bridge Lemma: $\mathrm{ECE}_B \leq \mathrm{CE}_{\mathrm{smooth}} + L/B$. Taking $B = O(\sqrt{m}/L)$:

$$\mathrm{ECE}_B = O(1/\sqrt{m}) \tag{86}$$

Comparison to nonparametric rate: The ratio is $(1/\sqrt{m})/(L\varepsilon/m)^{1/3} = (L\varepsilon)^{-1/3}/m^{1/6}$, which shrinks with m. For $m = 10{,}000$ and $L\varepsilon = 0.05$: ratio ≈ 0.05. Temperature scaling is $\sim$20$\times$ more data-efficient than nonparametric ECE estimation. □

F Verification Transfer Details

We provide the full proof of Proposition 32.

Step 1: Problem reduction. Since $\Delta_1(p)$ is known (from prior verification), estimating $\mathrm{ECE}(M_2) = \int |\Delta_2(p)|\, d\mu(p)$ reduces to estimating the shift $h(p) = \Delta_2(p) - \Delta_1(p)$, then computing $\int |\Delta_1(p) + \hat{h}(p)|\, d\mu(p)$.

Step 2: Lower bound (detection cost). Two-sample Le Cam between H_0: $\Delta_2 = \Delta_1$ and H_1: $\Delta_2 = \Delta_1 + h$ with $\|h\|_\infty = \delta_{\mathrm{ft}}$. From m_2 fresh samples near $p \approx 1 - \varepsilon_2$:

$$\mathrm{KL}(H_0^{m_2} \| H_1^{m_2}) \approx \frac{m_2 \delta_{\mathrm{ft}}^2}{\varepsilon_2} \tag{87}$$

Indistinguishable when $\mathrm{KL} \leq 1$, requiring $m_2 = \Omega(\varepsilon_2/\delta_{\mathrm{ft}}^2)$.

Step 3: Upper bound (estimation with transfer). Apply Theorem 9 to h instead of Δ_2. The histogram estimator for h has bias L_h/B and variance $\sqrt{\varepsilon_2 B/m_2}$. Optimizing: $B^* = (L_h^2 m_2/\varepsilon_2)^{1/3}$, giving:

$$\|\hat{h} - h\| = O\left(\left(\frac{L_h \varepsilon_2}{m_2}\right)^{1/3}\right) \tag{88}$$

By the triangle inequality:

$$\left|\mathrm{ECE}(M_2) - \widehat{\mathrm{ECE}}(M_2)\right| \leq \int |\hat{h}(p) - h(p)|\, d\mu(p) = O\left(\left(\frac{L_h \varepsilon_2}{m_2}\right)^{1/3}\right) \tag{89}$$

Setting this equal to δ_{ft}: $m_2 = O(L_h \varepsilon_2/\delta_{\mathrm{ft}}^3)$.

Step 4: When does transfer help? From-scratch: $m_2 = \Theta(L\varepsilon_2/\delta^3)$. With transfer: $m_2 = \Theta(L_h \varepsilon_2/\delta_{\mathrm{ft}}^3)$. Transfer helps when $\delta_{\mathrm{ft}} < \delta$ and $L_h \leq L$. With $\delta_{\mathrm{ft}} = \delta/10$, saving is 1000$\times$. Transfer provides no benefit when $\delta_{\mathrm{ft}} \geq \delta$. □

G Corrections: Correlation, Heterogeneity, Pre-Calibration

The core theorems assume i.i.d. labeled samples with independent Bernoulli noise. In practice, several deviations may arise.

Temporal correlation. If evaluation samples arrive sequentially and exhibit temporal dependence, the effective sample size is reduced by a factor depending on the mixing time. For β-mixing sequences with coefficient $\beta(k) \leq c \cdot \rho^k$, the effective sample size is $m_{\text{eff}} \approx m \cdot (1 - \rho)$. All bounds apply with m replaced by m_{eff}.

Domain heterogeneity. When the evaluation set spans multiple domains with different error rates, the per-domain verification floor (Fix 3) accounts for this heterogeneity. The overall ECE is a weighted average of domain-specific ECEs, and Jensen's inequality ensures the overall verification floor is no larger than the worst per-domain floor weighted by domain proportion.

Pre-calibration via RLHF. Models that have undergone reinforcement learning from human feedback may have artificially smooth calibration functions (small L), which reduces the verification tax. However, if the RLHF procedure oversmooths, the remaining miscalibration may be concentrated at specific confidence levels, violating the uniform Lipschitz assumption. Domain-specific Lipschitz estimates should be used in such cases.

H Numerical Sanity Checks

H.1 Sharp Le Cam Constants

For each ε, we numerically optimize the Le Cam lower bound over the separation parameter δ, using the exact Bernoulli KL divergence (not the quadratic approximation). We compute three versions: the Bretagnolle–Huber bound $R^* \geq (\delta/2)\exp(-\text{KL})$, the Pinsker bound $R^* \geq (\delta/2)(1-\sqrt{\text{KL}/2})$, and the exact bound via the Bernoulli likelihood ratio test. Table 3 reports the tightest constant for each ε.

Table 3: Sharp Le Cam constants $c_1(\varepsilon)$ and nonparametric estimation sample sizes. The Le Cam constant governs detection; the estimation column uses $m \geq L\varepsilon/\delta^3$ with $L = 1$, $\delta = 0.01$.

ε	$c_1(\varepsilon)$	m_{detect} (Le Cam)	m_{estimate} (nonparametric)
0.01	0.3780	15	10,000
0.02	0.3638	27	20,000
0.05	0.3474	61	50,000
0.10	0.3326	111	100,000
0.15	0.3206	155	150,000
0.20	0.3093	192	200,000
0.25	0.2982	223	250,000
0.30	0.2873	248	300,000
0.35	0.2759	267	350,000

The constant $c_1(\varepsilon)$ is monotonically decreasing in ε: smaller error rates yield tighter constants because the Bernoulli variance is smaller and the KL approximation is sharper. All values fall in $[0.28, 0.38]$, consistent with the theoretical prediction $c_1 \geq 1/(2e) \approx 0.184$ from the quadratic KL approximation. The exact constants are roughly $2\times$ larger, reflecting the tightness gained from exact Bernoulli KL computation.